SENS-D — NAFSI Hub Data Challenge

Proactive One Health Decision Support for Food Safety and Food Security

Primary decision-maker: Team-selected supply-chain actor (e.g., processor, distributor/logistics operator, retailer, or regional food bank)

Challenge question: How can a supply-chain actor (e.g., processor, distributor/logistics operator, retailer, or regional food bank) anticipate emerging food-safety risk and make proactive operational decisions that protect health while preserving food availability, supporting food security, and minimizing avoidable food loss?

1. Context

Foodborne risk does not originate at a single point in the food system. It emerges from interactions among human illness, animal and food production, processing environments, weather and environmental conditions, transportation, storage, and the communities receiving food. A One Health approach therefore requires teams to connect signals across human, animal/food, and environmental domains rather than analyze any single dataset in isolation.

Most food-safety actions occur after a positive test, outbreak signal, recall, or other adverse event is recognized. The SENS-D challenge asks teams to move upstream: use multiple data streams to identify changing risk conditions early enough to support a real operational decision. The anchor use case is Salmonella and other hazards in FSIS-regulated perishable foods, with poultry as a particularly relevant commodity, but teams may broaden the analysis when justified by the data.

Food-supply-chain decisions can create a direct tension between food safety and food security. Holding, discarding, delaying, rerouting, or changing the disposition of food may reduce potential exposure but can also increase food loss, disrupt supply, or reduce the amount of food available to communities experiencing food insecurity. A useful decision-support tool must therefore make risk visible without treating safety, food availability, and access as separate problems.

2. Challenge Overview

Your mission is to build a proactive, data-driven decision support tool for a supply-chain actor of your team's choosing—for example, a processor, distributor/logistics operator, retailer, or regional food bank. The tool should integrate human-health surveillance, food/animal production and regulatory data, environmental conditions, and food-security need to help the selected actor decide where to focus attention and how to act before a food-safety problem becomes a larger public-health event.

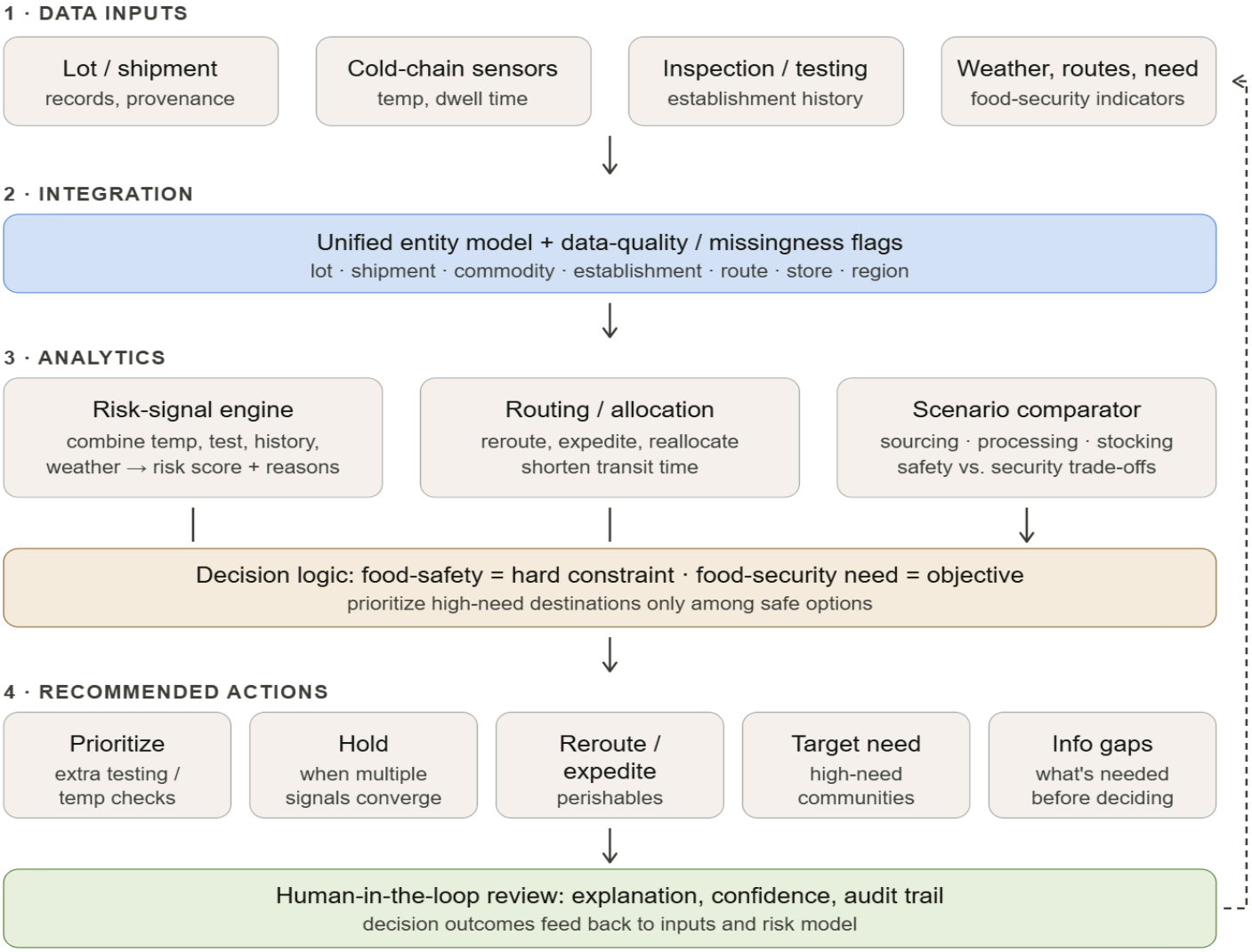
The challenge has five goals:

- Integrate required One Health datasets into a reproducible spatiotemporal data pipeline.
- Develop and validate a data-science model that detects or predicts elevated food-safety risk.
- Translate model outputs into concrete supply-chain decisions that balance food safety, food availability, and equitable access.
- Operationalize the decision logic in an interactive application developed with AWS Kiro.
- Demonstrate team science by combining disciplinary perspectives and incorporating additional datasets that materially improve the analysis or decision support.

Required decision context and actor selection

Teams must choose one primary supply-chain actor—such as a processor, distributor/logistics operator, retailer, or regional food bank—and clearly define that actor's operational role, decision authority, time horizon, and point in the supply chain. Teams should maintain the same primary actor throughout the challenge so that the analytics, decision logic, and interface are built around a coherent real-world use case. The final tool must support at least three decision actions appropriate to the selected actor, such as:

- Prioritize a lot, shipment, commodity, establishment, route, store, or region for additional review, testing, or temperature/quality checks.
- Place a lot, shipment, commodity, or other relevant unit on hold for human review when multiple risk signals converge.
- Expedite, reroute, reallocate, or otherwise adjust the movement of perishable food to reduce time in transit or avoid weather-related disruption while maintaining food-safety constraints.
- Prioritize destinations, markets, or communities where food-security need is high while maintaining food-safety constraints.
- Compare alternative processing, sourcing, routing, stocking, or allocation scenarios and explain the food-safety and food-security trade-offs.
- Identify what additional information is needed before a high-consequence decision is made.



3. Core Data and Data Integration Requirements

All teams must use the three core data streams below. Teams must also integrate at least two additional datasets of . Added datasets should improve prediction, interpretability, equity, operational relevance, or One Health completeness—not simply increase the number of variables.

Required source	One Health role	Suggested use	Access / organizer note
CDC PulseNet or organizer-provided PulseNet-derived extract	Human health / genomic surveillance	Emerging illness clusters, pathogen/subtype signals, temporal and geographic patterns	Direct PulseNet database access is limited to participating laboratories; organizers should provide a challenge-ready de-identified or aggregated extract.
USDA FSIS public datasets	Food / animal production / regulatory surveillance	Laboratory sampling, establishment information, Salmonella and other pathogen results, recalls and public-health alerts	Use FSIS sampling metadata carefully; FSIS warns that sampling or WGS identifiers alone are not sufficient to infer causal linkage to human illness.
NOAA / NCEI weather and climate data	Environment	Temperature, precipitation, extreme heat, storms, and other conditions relevant to cold-chain stress, transport disruption, and pathogen ecology	NOAA Climate Data Online provides API access; a free token is required for the CDO v2 API.

Additional datasets: required team contribution

Each team must add at least two external datasets, with at least one dataset that represents food security, population vulnerability, supply-chain operations, animal health, environmental exposure, or infrastructure.

Examples include:

- USDA Food Access Research Atlas, USDA Economic Research Service food-security indicators, Census ACS, or CDC/ATSDR Social Vulnerability Index.
- NCBI Pathogen Detection, state or local outbreak data, wastewater surveillance, or other public-health surveillance sources.
- Transportation networks, road closures, freight or travel-time data, cold-chain or temperature-monitoring data, or facility locations.
- Animal-health, agricultural production, land-use, water-quality, or environmental monitoring data.
- Actor-specific operational data if available and shareable, such as processor production/testing data, distributor inventory and routing data, retailer stocking/sales data, or food-bank inventory, demand, donation, distribution, or service-area data.

Teams must document provenance, licensing, spatial and temporal resolution, missingness, update frequency, and the specific decision value contributed by every dataset.

4. Tasks

All tasks are required. The tasks are progressive: each stage should feed the next, from One Health data integration to predictive modeling to an operational decision-support application.

Task 1 — Define the One Health system and build the integrated data layer

Start with the decision-maker—not the model. Select a primary supply-chain actor and define that actor's decisions, time horizon, geography, commodity scope, operational constraints, and acceptable actions. Then construct a One Health data model that connects the required datasets and your team-selected additions.

- Create a system map showing human, food/animal, environmental, supply-chain, and food-security pathways relevant to the decision.

- Define common spatial and temporal units and document how datasets are aligned, lagged, aggregated, or cross-walked.
- Create a data dictionary and provenance table.
- Identify known limitations and avoid treating temporal or genomic similarity as proof of outbreak causation.
- Define the target decision and the model outcome that will support that decision.

Expected output: One Health system diagram, data inventory/provenance table, harmonized analytic dataset, and a short decision statement describing who will use the tool, what decision they face, and when the decision must be made.

Task 2 — Develop and validate a proactive food-safety risk model

Build a data-science model that estimates near-term elevated risk at a decision-relevant unit such as commodity × region, establishment × time, or distribution corridor × time. Teams may use classification, anomaly detection, forecasting, spatial modeling, or an ensemble approach, but must justify the target and horizon.

- Create a transparent baseline model using a limited set of variables.
- Develop an integrated model using PulseNet-derived, FSIS, NOAA, and additional team-selected features.
- Use time-aware validation; avoid random train/test splits when they leak future information.
- Report discrimination and calibration metrics appropriate to the task, not accuracy alone.
- Quantify uncertainty and examine false-positive and false-negative consequences.
- Evaluate the marginal value of at least one added dataset or data domain.

Expected output: Reproducible modeling pipeline, performance metrics, calibration/uncertainty results, feature or signal interpretation, and a map or time series showing where/when elevated risk is identified.

Task 3 — Convert risk prediction into a food safety + food security decision model

Prediction is not decision support. Translate the risk model into explicit actions for the selected supply-chain actor and show how recommendations change when both food safety and food availability are considered.

- Define feasible actions such as inspect, hold for review, expedite, reroute, prioritize, or request additional information.
- Define operational constraints such as inventory, perishability, transport time, storage capacity, geography, and population need.
- Include at least one food-security or equity objective, such as preserving food volume, maintaining service to high-need communities, or minimizing disruption to vulnerable populations.
- Compare a simple rule-based or status-quo policy with the model-guided policy.
- Use scenario analysis or optimization to demonstrate trade-offs among risk reduction, food retained/distributed, travel or delay, and equitable access.
- Require human review for high-consequence decisions; the tool must support—not replace—regulatory and operational judgment.

Expected output: Decision logic or optimization formulation, at least three scenario comparisons, and quantitative measures showing both food-safety and food-security consequences.

Task 4 — Operationalize the decision support tool with AWS Kiro

Use AWS Kiro to translate the analytics into a working decision-support application. The model and decision logic should be visible to the user through an interface that supports action, not just visualization.

- Use Kiro Specs or equivalent structured prompts to define user requirements, technical design, and implementation tasks.
- Use Kiro steering files or project guidance to encode the selected actor, data definitions, decision rules, uncertainty language, and safety constraints consistently across the application.

- Build an interactive web application (React is recommended for consistency with prior NAFSI Hub challenges) that can be deployed from the team repository.
- At minimum, include: a current-risk view, a prioritized alert/attention queue, a scenario builder, a recommended-action panel, and an explanation/evidence panel.
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- Display uncertainty and the data signals supporting each recommendation. Do not label a food, establishment, or shipment “safe” or “unsafe” solely from the model.
- Export or precompute model outputs as reproducible CSV/JSON artifacts if a live backend is not practical. No additional AWS service is required beyond Kiro.

Expected output: Functional decision-support application, Kiro specification/prompt files, reproducible build instructions, and a documented link between model outputs and user-facing recommendations.

Task 5 — Demonstrate team science and the value of new data

This challenge is intentionally broader than a single-discipline modeling exercise. Teams should combine food safety, public health, data science, agriculture/environment, supply-chain, and/or user-design perspectives.

- Assign and document team roles; roles may overlap, but each team must show how different expertise shaped the solution.
- Integrate at least two additional datasets beyond the three required core sources.
- Run an ablation or comparison showing what changed when at least one new dataset or domain was added.
- Document at least one team decision that changed because of input from another discipline or stakeholder perspective.
- Identify the next dataset or partner that would most improve the tool if the project continued.

Expected output: One-page team science statement and a concise analysis of the added value of multi-source data integration.

5. Evaluation Criteria

Criterion	Weight	What judges will assess
Data science rigor and validation	25%	Sound target definition, preprocessing, time-aware validation, appropriate metrics, calibration/uncertainty, reproducibility, and comparison to a baseline.
Operational decision support	25%	Whether the tool changes or improves a real decision for the selected supply-chain actor; quality of scenarios, constraints, action logic, and food-safety/food-security trade-offs.
One Health and multi-source data integration	20%	Meaningful integration of human, food/animal, environmental, supply-chain, and food-security information; added datasets must contribute decision value.
Kiro implementation and usability	15%	Quality of Kiro specs/prompts/steering, working application, transparency of

		recommendations, user flow, and reproducible build.
Team science, reproducibility, and communication	15%	Cross-disciplinary collaboration, clear documentation, provenance, role contributions, transparent limitations, and effective report/presentation.

6. Questions to Address in the Final Report

1. Who is your decision-maker, what specific decision are you supporting, and what is the required decision horizon?

2. How does each core dataset contribute to the One Health picture? Where are the human, food/animal, environmental, and food-security signals?

3. Which additional datasets did your team add, why were they selected, and what changed after they were incorporated?

4. What outcome or risk signal did you model, and why is it appropriate for proactive decision support?

5. How does the integrated model perform relative to your baseline, including calibration and performance over time or across geography?

6. Which errors matter most for this use case? What are the consequences of false alarms versus missed risk?

7. Where did food safety and food security objectives conflict, and how did your decision logic or optimization approach handle the trade-off?

8. Walk through one realistic scenario from incoming signals to model output to recommended action. What evidence would the user see?

9. How did team science change the analysis, model, or tool design?

10. What can your tool support today, what should it never claim, and what data or validation would be required for real-world deployment?

7. Submission and Deliverables

Each team submission should include the following components in a public or organizer-accessible GitHub repository:

1. **Reproducible analysis notebooks/scripts:** All preprocessing, data integration, modeling, evaluation, and scenario analysis. Results should reproduce the outputs reported in the final report.
2. **Kiro development artifacts:** Kiro specs and/or prompt markdown, steering files or equivalent project guidance, and any hooks used for testing or documentation. Teams should disclose substantial manual changes to Kiro-generated code.
3. **Decision-support application:** Runnable web application and build/deployment instructions. React + GitHub Pages is recommended for consistency with the prior NAFSI Hub application challenge.
4. **Data dictionary and provenance file:** Source, access date, license, geography, time coverage, update frequency, transformations, and known limitations for each dataset.
5. **PDF report:** Methods, task outputs, responses to the questions above, model metrics, figures, decision scenarios, limitations, and team science statement.
6. **README and requirements:** Clear setup instructions and all dependencies needed to reproduce the analytic and application components.

The final report should reflect the team's own reasoning and documented experience. Any AI-assisted code or substantial AI-assisted content should be disclosed according to challenge rules. Teams should not update the repository after the official submission deadline.

8. Guardrails for Responsible Interpretation

- The challenge tool is a decision-support prototype, not a regulatory determination, outbreak declaration, or substitute for epidemiologic investigation.
- FSIS sampling, WGS identifiers, and allele codes can inform surveillance and risk prioritization but should not be used alone to claim causal linkage to human illness.
- Recommendations should expose uncertainty, provenance, and the signals driving the recommendation.
- High-consequence actions should include a human-review step and identify what additional evidence would be needed.
- Teams should test whether recommendations systematically disadvantage high-need communities or reduce food access in ways not justified by the risk evidence.

9. Organizer Notes and Source Links

PulseNet access note: CDC describes PulseNet as a national laboratory network and database used by public-health and food-regulatory laboratories to detect outbreaks. For a student challenge, organizers should provide a challenge-ready PulseNet-derived file or approved aggregate rather than assume direct database access. If that is not feasible, a public CDC human-illness dataset can be substituted while retaining the requirement to integrate a human-health surveillance signal.

FSIS data note: FSIS publishes establishment-specific laboratory sampling data and a Recall/Public Health Alert API. Its laboratory sampling documentation explicitly cautions that the posted data are not sufficient by themselves to establish associations among samples or causal links to human illness.

NOAA data note: NOAA/NCEI Climate Data Online provides programmatic access to station and location data through the CDO v2 API. Access is free but requires a token.

Kiro note: Kiro supports structured specifications, steering files, hooks, and agentic development workflows that can be used to turn the team's decision logic into a reproducible application.

- CDC PulseNet
- USDA FSIS Data Sets and Visualizations
- USDA FSIS Laboratory Sampling Data
- USDA FSIS Developer Resources / Recall API
- NOAA/NCEI Climate Data Online API
- Kiro Documentation